Meheri Monir

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Research Interests

Broadly interested in machine learning and AI for high-impact, real-world decision-making, with a growing interest in interpretability, uncertainty estimation and trustworthy AI for high-stakes domains. Open to exploring new directions in graph-based learning, human-centered AI and applications beyond my current healthcare-focused work.

Education

International Islamic University Chittagong — *Chattogram, Bangladesh*
Bachelor of Science in Computer Science & Engineering — *Expected September 2026*
CGPA: 3.918 / 4.00 (Thesis defense pending)

Government City College Chattogram — *Chattogram, Bangladesh*
Higher Secondary Certificate (HSC) — *2021*
GPA: 5.00 / 5.00

Bakalia Government High School — *Chattogram, Bangladesh*
Secondary School Certificate (SSC) — *2019*
GPA: 5.00 / 5.00

Experience

- **Research Intern, EliteLab** (July 2026 – Present): Assigned to ongoing research projects under lab researchers, contributing to experiment design, literature review and data analysis, with co-authorship on resulting papers
- **Undergraduate Teaching Assistant - Dept. of CSE, IIUC - Computer Networks & Image Processing**: Conducted lab sessions, graded assignments/exams and mentored an outstanding project team in writing a paper

Publications

- **Dengue Prediction Using Google Search Trends: A Machine Learning Approach** — COMPAS 2025, Published, 2025
- **Predicting Food Prices Using Machine Learning and SHAP** — QPAIN 2025, Published, 2025
- **CCA-HyGNN: A Cross-Context Attention Hypergraph Neural Network for Multi-Drug Side Effect Prediction** — ICISET, Under Review, 2026
- **HARP-TS: Confidence-Driven Two-Timescale Parameter Adaptation for ACO Routing in Hierarchical Vehicular Delivery Networks** — ICISET, Under Review, 2026
- **Attention-Based Interpretable Pan-Cancer Classification on TCGA Multi-Omics Data** — Manuscript in Preparation, 2026
- **RMSpropW: Bias-Corrected RMSprop with Decoupled Weight Decay for Imbalanced Classification** — Manuscript in Preparation, 2026
- **A Targeted Lesion Fix and a Systematic Diagnosis of Lesion-Grade Coupling in Diabetic Retinopathy Grading** — Manuscript in Preparation, 2026

Undergraduate Thesis

HealGNN: Predicting Disease Risk from Environmental Features and Simulating Goal-Directed Counterfactual Policy Interventions via Spatio-Temporal Graph Neural Networks

HealGNN is a spatiotemporal graph neural network that predicts chronic disease prevalence across Ohio's Core Based Statistical Areas using environmental and land-cover data. It combines a GAT branch with a Spatial Transformer branch via gated fusion to capture both local and global spatial patterns over time. A companion module, Urban Surgeon, then uses the trained model to simulate targeted environmental interventions that could reduce predicted disease risk.

Projects

- Passive Attack Detection in Computer Security
- Career Recommender System
- Image Reconstruction using Numerical Methods
- University Forum & Marketplace
- LAN Multiplayer Game Hub
- Object Classification Car
- Donation Tracker DBMS
- Mini apps & games

Technical Skills

Languages	Python, PHP, JavaScript, Dart, C, C++, C#
ML, DL & Computer Vision	PyTorch, TensorFlow, PyTorch Geometric, Scikit-learn, NumPy, Pandas, SHAP, scikit-image, Albumentations, torchvision
Web, Mobile & Database	HTML/CSS, Tailwind CSS, Flutter, MySQL, phpMyAdmin

Achievements & Extracurricular Activities

- Multiple 1st/2nd place finishes across regional chess and art competitions
- Participated in datathons and programming contests; regular attendee of university CSE fests

Languages

Bangla	Native
English	Fluent